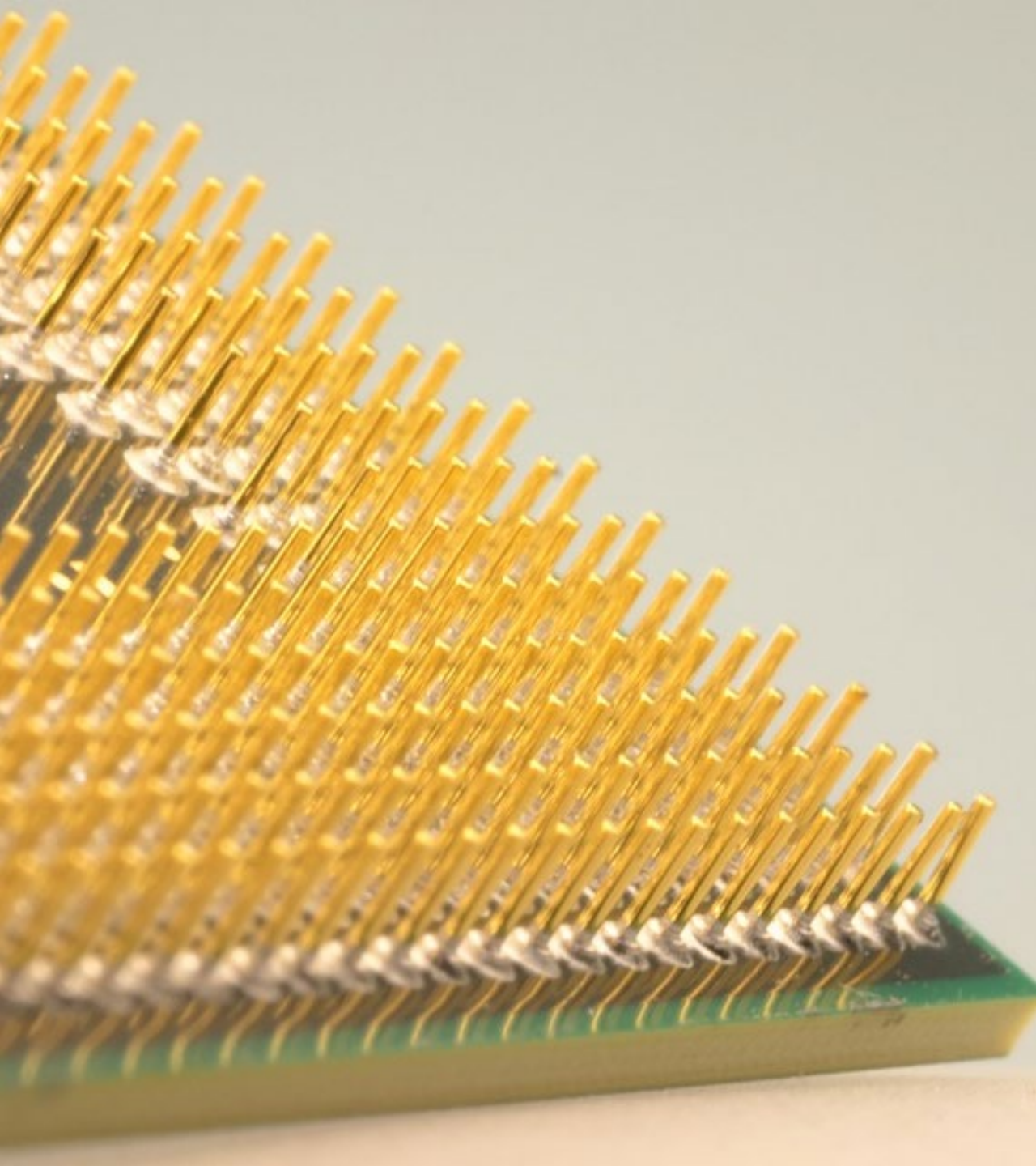




ME 300:
Engineering Thermodynamics

Statements 2-4 of the Second Law

Second Law: Statement 1

1A: $\bar{W} \rightarrow Q$ completely, $Q \rightarrow \bar{W}$ completely & continuously

1B: (K-P) impossible to create cycle with heat from a single reservoir where $\bar{W}_{net} = Q_H \Rightarrow Q_H = \bar{W}_{net} + Q_L$



I put a hot coffee on the kitchen table – what happens?

- ☒ A. Cools down
- ☐ B. Stays the same temperature
- ☐ C. Heats up

Second Law: Statement 2

Clausius original statement: heat flows spontaneously from T_H to T_L ,
and it requires work to have heat flow
from T_L to T_H

→ spontaneous, processes have preferred direction, move towards
equilibrium

→ refrigeration

Second Law: Statement 3

Clausius entropy statement: the entropy of a system and its surroundings increases, or in the limit of a reversible process, stays constant.

↳ entropy: $S \sim [J/K]$

$$S = [J/K]$$
$$\bar{S} = [J/mol \cdot K]$$

Second Law: Statement 4

Equilibrium is achieved when entropy is maximized

for a simple system at constant u and v .

↑
internal energy
 $[J/kg]$

↑
mass-specific
volume
 $[m^3/kg]$